MITHIL K GOWDA

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Professional Summary

Computer Science graduate with experience in software engineering, secure application development, distributed systems, machine learning, backend engineering, and intelligent system design. Project Intern at ISRO–LEOS with hands-on experience developing scalable backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized research focused on AI-driven intelligent systems, distributed architectures, and machine learning. Passionate about software quality, algorithmic problem solving, system design, and evaluating AI-generated technical solutions. Seeking opportunities in Software Engineering, AI Research, Computer Science Research, and Intelligent Systems.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru 2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

Programming: Python, Java, JavaScript, SQL, Bash

Software Engineering: Object-Oriented Programming, Data Structures, Algorithms, Software Design, Design Patterns, Secure Software Development

Backend Development: Flask, REST APIs, API Design, Backend Architecture

Machine Learning: XGBoost, CNN-LSTM, Deep Q-Network (DQN), Supervised Learning, Reinforcement Learning, Predictive Modeling

Computer Science: Distributed Systems, Database Design, System Design, Computer Architecture Fundamentals, Cloud Fundamentals

Databases: MySQL, MongoDB

Tools: Git, GitHub, VS Code, Postman

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL following Secure SDLC principles supporting enterprise software systems.
- Implemented secure authentication, authorization, API security, and database protection mechanisms emphasizing secure application architecture.
- Developed scalable backend services supporting engineering analytics, distributed application workflows, intelligent data processing, API development, and enterprise software reliability.
- Engineered modular backend architectures emphasizing software quality, maintainability, scalability, performance optimization, and clean software design principles.

Research Projects Software Engineering

A Multi-Agent Intelligent Decision Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven intelligent analytics framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good*.
- Designed a distributed multi-agent intelligent system integrating machine learning, adaptive decision making, reinforcement learning, distributed coordination, and scalable software architecture.

- Developed intelligent machine learning models using XGBoost, CNN-LSTM, and Deep Q-Network (DQN), applying feature engineering, model evaluation, predictive analytics, and distributed learning.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise security scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed an intelligent software platform integrating trust scoring, predictive analytics, authentication, decision-support systems, and distributed transaction management.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, API security, secure authentication, and enterprise-grade transaction validation.

C^3T-STAVP Cryptographic Framework (Ongoing Research)

- Designing an intelligent data-driven framework integrating Character Chaffing & Transposition Technology (C^3T), Secure Transaction Authorization & Verification Protocol (STAVP), and adaptive trust validation.
- Researching intelligent optimization, semantic data representation, distributed systems, privacy-preserving computation, and resilient software architectures.
- Developing intelligent cryptographic security mechanisms supporting secure cloud architectures, enterprise software systems, adaptive trust management, and resilient distributed applications.

Professional Certifications

- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Ethical Hacker – Cisco Networking Academy
- Systems and Usable Security (Elite) – NPTEL
- Google Cloud Computing – NPTEL
- Business Intelligence and Analytics – NPTEL
- IBM Front-End Developer Specialization – IBM
- Introduction to Cyber Security – IGNOU (SWAYAM)
- Introduction to Internet of Things (Silver Medal) – NPTEL

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO - LEOS).
- First Author of internationally recognized research focused on AI-driven intelligent systems, distributed software architectures, machine learning, and intelligent decision-support systems.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.